

Proof & Equation Verification Report — gbc_downside_main.tex

Date: 2026-08-03 **Scope:** Every displayed equation and all four formal results (two lemmas, two propositions) in the Methods section (lines 376–1150). No new theorems or proofs appear after line 1150; later sections only reference these results.

Verification method (five independent passes):

1. My own line-by-line re-derivation + numerical checks in Python (SciPy).
2. Independent fresh-context auditor A (“re-derive everything, state missing hypotheses”).
3. Independent fresh-context auditor B (adversarial: “try to break each proof,” with Monte-Carlo reproductions).
4. **ChatGPT** (Plus, High reasoning) — driven in-browser, section-by-section, ran its own code.
5. **Gemini** (Pro, Extended thinking) — driven in-browser, section-by-section.

The five passes converge. Headline finding: **no result is mathematically false, and nothing is hallucinated**. The underlying mathematics is sound and the numbers reproduce exactly. What the audit found is a set of *missing hypotheses* and *terse steps* that a top journal referee would flag, plus **one genuine factual error in a prose remark** and **one overclaim about continuity**. All are fixable without changing any result.

1. Consensus verdict table

#	Result	Me	Auditor A	Auditor B	ChatGPT	Gemini	Consensus
L1	Pinball regret identity	Correct	Correct-but-terse	Cannot break	GAP (atom/derivative)	Correct-but-terse	Correct; justify the derivative step; add finite-mean hypothesis
P1	Split-conformal validity	Correct	Correct-but-terse (missing hyp.)	Broke upper bound with ties	ERROR (ties + endpoint)	GAP (no-ties)	Correct given a no-ties/continuity hypothesis — which is currently missing
P2	Tail wedge (kurtosis)	Correct	Correct (remark has error)	Partial ($\tau^* < 1/2$ not shown)	Terse (τ^* fine via min)	GAP ($\tau^* < 1/2$)	Result correct; add one line that $\tau^* \in (0, 1/2)$ by intersecting with $1/2$
L2	Generative sampler	Correct	Correct-but-terse	Cannot break	Correct-but-terse	Correct	Correct; make the inverse-of-inverse step explicit

#	Result	Me	Auditor A	Auditor B	ChatGPT	Gemini	Consensus
Eq	EVT/GPD splice (eq:evt)	Correct; algebra; splice caveat	Correct; splice caveat	Correct; splice caveat	ERROR <i>for the conditional splice; algebra correct</i>	ERROR (continuity flawed)	Algebra correct; “splices continuously” overclaims — see §4
Rk	Crossover remark (rem:crossover)	—	ERROR	(numeric)	ERROR ($\tau \approx 0.0179$)	ERROR ($\tau \approx 0.0179$)	Factual error: fix the wording — see §3
Eq	Pinball, GARCH-t, IQN, Gibbs, score	Correct	Correct	Correct	—	—	All correct

Numerical claims that were re-checked and reproduce exactly: unit-variance t_3 quantiles (5%: -1.36 vs normal -1.64 ; 1%: -2.62 vs normal -2.33); the pinball regret identity vs Monte Carlo; the t_3 -vs-normal crossover level (all three code-running checkers independently returned $\tau \approx 0.01794$).

2. The four proofs — detailed findings

L1 — Pinball regret identity \label{lem:regret}

Correct. $dS/dq = F(q) - \tau$ and integrating gives the identity; the quadratic sandwich follows from $m \leq f \leq M$. Two things a referee will demand:

- The derivative step is written as “by dominated convergence,” but ρ_τ is not differentiable at 0. The correct justification is the Lipschitz bound $|\rho_\tau(Z-q) - \rho_\tau(Z-q')| \leq \max(\tau, 1-\tau) |q-q'|$ (dominated *difference quotients*). At an atom the one-sided derivatives differ: $S'_-(q)=F(q^-)-\tau$, $S'_+(q)=F(q)-\tau$; since the result is then integrated, atoms are a null set, so **the identity itself needs no continuity assumption** — worth stating as a strength.
- The quadratic bound’s step $F(u)-\tau = F(u)-F(q_F)$ needs $F(q_F)=\tau$, which is *supplied* by the density hypothesis (absolute continuity $\Rightarrow F$ continuous). One clause fixes it.
- Add the standing hypothesis $E|Z| < \infty$ so $S(q)$ is finite.
- Fix the nonnegativity wording: as written it only literally covers $q > q_F$; state the $q < q_F$ case (reversed orientation) too.

P1 — Split-conformal validity \label{prop:conformal}

Correct, but a required hypothesis is missing — this is the most important fix. Under exchangeability and a.s. distinct residuals, the rank of e_{n+1} is uniform on $\{1, \dots, n+1\}$, coverage $= [(n+1)\tau]/(n+1) \in [\tau, \tau+1/(n+1))$.

- **Missing no-ties/continuity assumption.** Auditor B and both external models broke the *upper* bound by forcing ties (with identical residuals coverage $\rightarrow 1$). Add “continuous error distribution (a.s. distinct residuals), or randomized tie-breaking.”

- **Endpoint caveat (ChatGPT + Auditor B):** if $\lceil (n+1)\tau \rceil = n+1$ (τ near 1), the order statistic of n calibration points is undefined; adopt the convention $c_\tau = +\infty$ there. Relevant because the shift is applied per level across the whole curve, including upper levels.
- The upper bound is actually **strict** ($< \tau + 1/(n+1)$); keeping $\leq$ is true but never attained. Coverage is **marginal**, not conditional on s — worth one clause given the pooled-state setup.

P2 — Tail wedge under kurtosis misspecification \label{prop:wedge}

Correct. Regular variation of the unit-variance t_v tail gives $Q_v(\tau) = -(c_v/\tau)^{1/v}(1+o(1))$; the unit-variance rescaling changes only the constant c_v , not the tail index v , so the ratio diverges and the ordering holds for small τ .

- On $\tau^* \in (0, \frac{1}{2})$: the asymptotics give ordering below *some* $\delta > 0$. The proposition only claims *existence* of a τ^* in $(0, \frac{1}{2})$, which follows immediately by taking $\tau^* < \min(\delta, \frac{1}{2})$ (ChatGPT’s nuance). Gemini/Auditor B flagged this as a gap because the proof doesn’t *locate* the crossover — but locating it isn’t claimed. Add the one-line $\min(\cdot, \frac{1}{2})$ remark and the point is airtight.
- State explicitly $v_1, v_2 > 2$ (finite variance) and cite the RV quantile-inversion (de Haan-Ferreira / Karamata).

L2 — Generative sampler validity \label{lem:sampler}

Correct. Standard inverse-transform. Two terse spots to expand: (i) $P(\tau \leq a) = a$ because $\tau \sim U(0, 1)$ (unstated); (ii) name the step that the generalized inverse of a nondecreasing left-continuous quantile function returns the function itself (the Galois “inverse-of-inverse” duality) — that last step *is* the content and is currently hand-waved as “which equals that supremum.” Left-continuity + monotonicity are jointly sufficient; no error.

3. The one genuine error: the crossover remark \label{rem:crossover}

The remark states the deployable levels “ $\tau \leq 0.025$ sit below the empirical crossover.” This is **numerically false**. All three code-running checkers put the t_3 -vs-normal quantile crossover at $\tau \approx 0.0179$ (1.79%). Therefore:

- $\tau = 0.01$ is below the crossover (fat tail more extreme) $\square$
- $\tau = 0.02$ and $\tau = 0.025$ are **above** it — there the unit-variance t_3 quantile is *less* extreme than the normal (at 2.5%: -1.837 vs -1.960).

The rhetorical point (that whole-distribution/CRPS scoring can rank a fat-tail model *below* a thin-tail one at these levels) is stated for exactly the level where it does not hold at 2.5%. **Fix:** reword to “the deepest deployable level ($\tau = 0.01$) sits below the crossover ($\approx 1.79\%$), while $\tau = 0.025$ straddles it.” The surrounding argument is unaffected.

4. The one overclaim: EVT “splices continuously” (eq:evt)

The **algebra is correct**: solving $\tau = p_\circ(1+\xi(x-u)/\beta)^{-1/\xi}$ gives $q_{\text{EVT}}(\tau) = -[u + (\beta/\xi)((\tau/p_\circ)^{-\xi} - 1)]$, and $q_{\text{EVT}}(p_\circ) = -u$. So the EVT curve is continuous *at its own endpoint*. But the paper says it “splices continuously into the Stage-2 curve,” and the Stage-2 body quantile at $p_\circ$ is the **state-conditional** $\tilde{q}(p_\circ|s)$, whereas u is fixed at the **unconditional** empirical loss quantile. These coincide only if $u = -\tilde{q}(p_\circ|s)$ for every s , which generally fails. **Fix:** either (a) anchor the threshold at the body model’s own $p_\circ$ -quantile, $u(s) = -\tilde{q}(p_\circ|s)$, or (b) downgrade the wording to “continuous at the EVT endpoint; approximately continuous with

the body up to the gap $-u - \tilde{q}(p_0|s)$.” Add the standing GPD hypotheses $\beta > 0$, $1 + \xi(x-u)/\beta > 0$, and the $\xi \rightarrow 0$ (exponential) limit.

5. Everything else checks out

- eq:pinball (check loss), eq:garch (GARCH-t recursion), eq:shapeERM, eq:hybrid, eq:iqn (cosine embedding — verbatim Dabney 2018 IQN), eq:gibbs (Gibbs/generalized-Bayes posterior), and eq:score (windowed excess-kurtosis / skewness moments): **all correct**.
 - The Gibbs ω one-step rescaling rule (posterior variance $\propto 1/\omega$, so $\omega \leftarrow \omega \cdot (\text{SD}_{\text{post}}/\text{se}_{\text{boot}})^2$ lands the SD in one step) was numerically confirmed and is honestly hedged to the approximately-Gaussian regime.
-

6. How top journals want this structured (and where to aim)

Based on the current author guidelines (2024–2026) of Annals of Statistics, JASA T&M, JRSS-B, Biometrika, Journal of Econometrics, and Journal of Financial Econometrics:

- **This is not an Annals-of-Statistics paper.** AoS wants deep theorems *as the payload*; four elementary (though correct and useful) formal results wrapped around a $\sim 70\%$ -empirical risk-forecasting study reads as “thin theory + application” there. The natural homes are **Journal of Financial Econometrics** or **Journal of Econometrics** (best fit for a VaR/ES downside-risk paper with strong empirics and supporting theory), or a **sharpened JASA Theory & Methods** version if the results are recast as methodological guarantees. Your own in-file FORMAT NOTE already targets JFEC/JBF/Quantitative Finance, which is consistent.
- **Proof placement:** the dominant convention (explicit in JRSS-B: “development of mathematical expressions... presented in appendixes, only the relevant equations in the body”) is to **state each result in the main text next to its assumptions, and defer full proofs to an appendix or online supplement**. Keep at most a one-line proof sketch inline for the headline result. $\rightarrow$ I have expanded the proofs and moved the rigorous versions to a new **Appendix B (Proofs)**, leaving compact statements + sketches in the Methods body.
- **Code:** never printed in the paper. JRSS-B and JASA now effectively *require* a public replication repository with a **Zenodo DOI cited in the main text**, plus a Data/Code Availability Statement; JFEC runs a replicability review as a condition of acceptance. Your algorithm boxes stay; add a one-paragraph availability statement pointing at the repo+DOI.
- **Figures/charts:** normal even in theory-forward papers, but *confirmatory and compact* — keep 1–4 core figures/tables in the body (the pipeline diagram and the frontier/decile results), push extended simulation grids and robustness tables to a supplement. A theory-forward paper is $\sim 10\text{--}25\%$ empirical; a methods paper $40\text{--}70\%$. At $\sim 70\%$ empirical you are squarely a *methods/econometrics* paper, which is fine for JFEC/JoE and argues against trimming the empirics to chase a pure-theory venue.

Sources: imstat.org (AoS manuscript prep & supplement instructions); academic.oup.com/jrsssb, /biomet, /jfec (general instructions); tandfonline.com JASA reproducibility (2024); sciencedirect.com Journal of Econometrics guide for authors.

7. Bottom line

Nothing in the paper’s mathematics is invented or wrong. Priority fixes before submission, in order:

1. **Reword `rem:crossover`** — the “ $\tau \leq 0.025$ below the crossover” claim is false; crossover $\approx 1.79\%$. (*genuine error*)
2. **Add the no-ties/continuity hypothesis to `prop:conformal`** (and the $\lceil (n+1)\tau \rceil = n+1$ end-point convention). (*missing hypothesis that breaks the stated upper bound*)
3. **Soften/repair the EVT “splices continuously” claim** (unconditional u vs state-conditional body). (*overclaim*)
4. **Expand the two terse proofs** (`lem:regret` derivative justification + finite-mean; `lem:sampler` inverse-of-inverse step) and **add the $\tau^* < \min(6, \frac{1}{2})$ line** to `prop:wedge`.
5. **Move full proofs to Appendix B**, keep statements + sketches inline; add a code/data availability statement.

Items 1-5 are implemented in the companion edit to `gbc_downside_main.tex` (see the new Appendix B and the inline fixes).